

Meeting IV: Intro to Category Theory Pt. 3

Topoi: The Categorical Analysis of Logic

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Recap

In part 1, we covered the definitions of a quiver, path, and category.

Intuition

A **category** is a quiver equipped with an associative, unital path reduction operation.

- ▶ The operation is expressible via a binary composition with identities.

In part 2, we covered diagrams as quiver homomorphisms to a category. We used them to describe the universal property of a categorical product.

Intuition

A **product** is a terminal object in the category of cones of an *arrowless* diagram.

We also briefly talked about isos, duality, and monics.

Goals for today

By the end of this meeting, we should understand:

- ▶ limit ($\text{colimit}^{\text{op}}$)
 - terminal ($\text{initial}^{\text{op}}$) object in a certain full subcategory of cones ($\text{cocones}^{\text{op}}$)
 - examples:
 - ▶ equalizer ($\text{coequalizer}^{\text{op}}$)
 - ▶ pullback ($\text{pushout}^{\text{op}}$)
- ▶ subobject classifier
 - subobject
 - ▶ isomorphism class of subterminals in a slice category
 - characteristic arrow
 - ▶ internal description of subobjects

Preliminaries

Terminal Object: Review

Let $\mathcal{C}$ be a category. Let T be a $\mathcal{C}$ -object.

Definition: Review

- ▶ T is **terminal** if for all $\mathcal{C}$ -objects X , there is a unique $\mathcal{C}$ -arrow f such that:
 - $\text{dom}(f) = X$
 - $\text{cod}(f) = T$
- ▶ In English: every object admits a unique arrow to T .
 - We may refer to this map $l_v : X \rightarrow T$ as the comparison map.

$$X \dashrightarrow_{l_v} T$$

We shall individually examine uniqueness and existence.

Subterminal: Definition

Let $\mathcal{C}$ be a category. Let U be a $\mathcal{C}$ -object.

Definition

- ▶ U is **subterminal** if for all $\mathcal{C}$ -objects X , there is at most one $\mathcal{C}$ -arrow f such that:
 - $\text{dom}(f) = X$
 - $\text{cod}(f) = U$
- ▶ In English: parallel maps to U coincide.

Intuition

- ▶ A subterminal satisfies the uniqueness aspect of a terminal.
- ▶ In a preorder, every object is subterminal.

Monomorphism: Review

Definition: Review

An arrow $f : a \rightarrow b$ is called **monic** if it is left-cancellable: $f \circ g = f \circ h$ implies $g = h$.

$$\begin{array}{ccc} x & \xrightarrow{h} & a \\ g \downarrow & & \downarrow f \\ a & \xrightarrow{f} & b \end{array} \text{ commutes implies } x \begin{array}{c} \xrightarrow{g} \\ \xleftarrow{h} \end{array} a \xrightarrow{f} b \text{ commutes}$$

$a \longleftarrow b \qquad a \rightrightarrows b$

Examples

- ▶ In **Set**, monomorphisms are precisely injective functions.
- ▶ In a preorder, every arrow is monic.

Subterminal Object: Equivalent Characterization

Let $\mathcal{C}$ be a category with a terminal object T . Let U be a $\mathcal{C}$ -object.

Lemma

U is subterminal iff the (unique) comparison map $U \xrightarrow{l_U} T$ is monic.

$$\begin{array}{c} X \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} U \end{array} \quad \begin{array}{c} \text{reverse} \\ \text{direction:} \end{array} \quad \begin{array}{c} X \begin{array}{c} \xrightarrow{l_U \circ f} \\ \xrightarrow{l_U \circ g} \end{array} T \end{array} \quad \begin{array}{c} \Rightarrow \\ \Rightarrow \end{array} \quad \begin{array}{c} l_U \circ f = l_U \circ g \\ f = g \end{array}$$

Intuition

- ▶ A subterminal object is like a “subset” of a terminal object (if it exists).
 - In **Set**, a subterminal is either the empty set or a singleton.

Weak Terminal Object: Definition

Let $\mathcal{C}$ be a category. Let W be a $\mathcal{C}$ -object.

Definition

- ▶ W is **weakly terminal** if for all $\mathcal{C}$ -objects X , there is a $\mathcal{C}$ -arrow f such that:
 - $\text{dom}(f) = X$
 - $\text{cod}(f) = W$

Intuition

- ▶ A weak terminal satisfies the existence aspect of a terminal.
- ▶ In **Set**, any non-empty set is weakly terminal.

Terminal Object: Revisited

Lemma

A $\mathcal{C}$ -object is terminal iff it is both subterminal and weakly terminal.

Intuition

Checking one criterion may often be easier than the other criterion.

Subcategory: Review

Let $\mathcal{C}' = (O', A', \text{dom}', \text{cod}', \circ', 1')$ and $\mathcal{C} = (O, A, \text{dom}, \text{cod}, \circ, 1)$ be categories.

Definition

$\mathcal{C}'$ is a **subcategory** of $\mathcal{C}$ if:

- ▶ (The underlying quiver of) $\mathcal{C}'$ is a **subquiver** of (the underlying quiver of) $\mathcal{C}$:
 - $O' \subseteq O$ and $A' \subseteq A$
 - $\text{dom}' = \text{dom}|_{A'}$ and $\text{cod}' = \text{cod}|_{A'}$
- ▶ The path reduction operations of $\mathcal{C}'$ are completely determined from those in $\mathcal{C}$:
 - $\circ' = \circ|_{A'_{\text{dom}} \times_{\text{cod}} A'}$
 - $1' = 1|_{O'}$

Intuition

The “inclusion” $\mathcal{C}' \subseteq \mathcal{C}$ is a quiver homomorphism that preserves path reduction.

Subcategory: Review

Let $\mathcal{C}' = (O', A', \text{dom}', \text{cod}', \circ', 1')$ and $\mathcal{C} = (O, A, \text{dom}, \text{cod}, \circ, 1)$ be categories.

Diagrammatic Intuition

$\mathcal{C}'$ is a subcategory of $\mathcal{C}$ iff $O' \subseteq O$ and $A' \subseteq A$ and the following diagrams commute:

$$\begin{array}{ccc} A' \hookrightarrow A & O' \hookrightarrow O & A' \hookrightarrow A \\ \text{dom}' \downarrow & \downarrow & \downarrow \text{cod}' \\ O' \hookrightarrow O & A' \hookrightarrow A & O' \hookrightarrow O \end{array}$$

$$\begin{array}{ccc} (A'_{\text{dom}' \times_{\text{cod}'}} A') \hookrightarrow (A_{\text{dom} \times_{\text{cod}}} A) \\ \circ' \downarrow & & \downarrow \circ \\ A' \hookrightarrow A \end{array}$$

Monics in a Subcategory

Let $\mathcal{C}'$ be a subcategory of $\mathcal{C}$. Let $f : A \rightarrow B$ be a $\mathcal{C}'$ -arrow.

Lemma

If f is monic in $\mathcal{C}$, then f is monic in $\mathcal{C}'$.

Notes

- ▶ The inclusion $\mathcal{C}' \subseteq \mathcal{C}$ “reflects” being a monic.
- ▶ The inclusion need not “preserve” being a monic.
 - But some inclusions do preserve monics!

Full Subcategory: Definition

Let $\mathcal{C}' = (O', A', \text{dom}', \text{cod}', \circ', 1')$ be a subcategory of $\mathcal{C} = (O, A, \text{dom}, \text{cod}, \circ, 1)$.

Definition

$\mathcal{C}'$ is a **full** subcategory of $\mathcal{C}$ if for all $\mathcal{C}'$ -objects X and Y :

$$\mathcal{C}'(X, Y) = \mathcal{C}(X, Y)$$

Reminder

The notation $\mathcal{C}(X, Y)$ stands for the “hom-collection” of $\mathcal{C}$ -arrows from X to Y :

$$\mathcal{C}(X, Y) = \{f \in A : \text{dom}(f) = X \wedge \text{cod}(f) = Y\}$$

Full Subcategory: Examples

Finset is a full subcategory of **Set**.

$$\{0\} \xrightarrow[\{\}]{\{\langle 0,0 \rangle\}} \{0\}$$

Set is a non-full subcategory **Rel**.

$$\langle A, B, R \rangle \text{ s.t. } R \subseteq A \times B \quad \therefore A \rightarrow B \text{ in Rel}$$

Note

Rel is the category with sets as objects and binary relations as arrows.

Limits and Colimits

Products: Review

Let $\mathcal{C}$ be a category and $D = (D_0, \emptyset)$ an arrowless diagram in $\mathcal{C}$ of shape $Q = (V, \emptyset, !, !)$.

Definition: Review

- ▶ A **cone** $(x, \{f_v\}_{v \in V})$ in $\mathcal{C}$ for the (arrowless) diagram D is:
 - a $\mathcal{C}$ -object x , together with
 - a $\mathcal{C}$ -arrow $f_v : x \rightarrow D_0(v)$ for each vertex $v \in V$.
- ▶ A **cone morphism** from $(x, \{f_v\}_{v \in V})$ to $(y, \{g_v\}_{v \in V})$ is an arrow $h : x \rightarrow y$ where:
 - $f_v = g_v \circ h \quad \forall v \in V$
- ▶ A **product** is a terminal object in the category of cones of an *arrowless* diagram.

What concepts do we get by allowing arrows in the diagram?

Cone: Definition

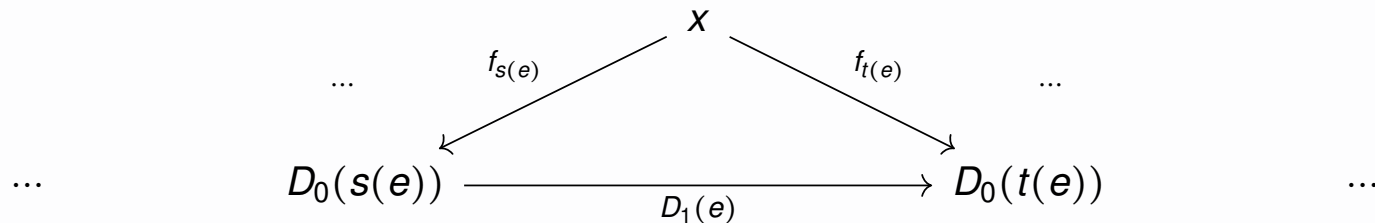
Let $D = (D_0, D_1)$ be a diagram in $\mathcal{C}$ of shape $Q = (V, E, s, t)$.

Let $C = (x, \{f_v\}_{v \in V})$ be a cone for the associated arrowless diagram $(D_0, \emptyset)$.

Definition

C is a **cone** for the diagram D if for each edge $e \in E$:

$$f_{t(e)} = D_1(e) \circ f_{s(e)}$$



Cocone: Definition

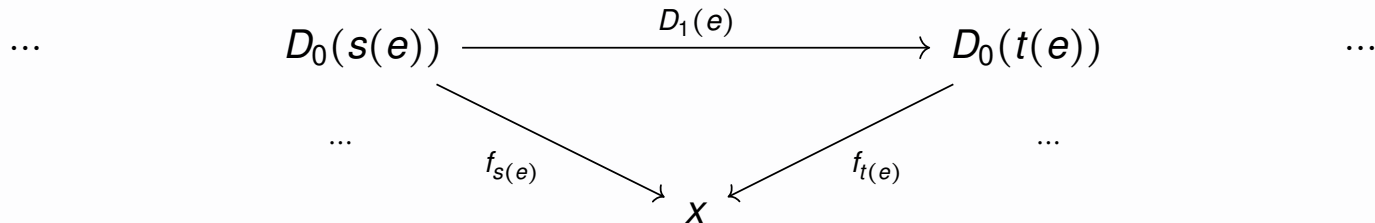
Let $D = (D_0, D_1)$ be a diagram in $\mathcal{C}$ of shape $Q = (V, E, s, t)$.

Let $C = (X, \{f_v\}_{v \in V})$ be a cocone for the associated arrowless diagram $(D_0, \emptyset)$.

Definition

C is a **cocone** for the diagram D if for each edge $e \in E$:

$$f_{s(e)} = f_{t(e)} \circ D_1(e)$$



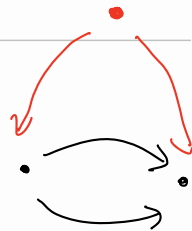
Cone: Does It Commute?

Let $D = (D_0, D_1)$ be a diagram of shape $Q = (V, E, s, t)$.

Let $C = (x, \{f_v\}_{v \in V})$ be a cone for the diagram D .

Consider the augmented shape $Q' = (V + \{x\}, E + V, s + i_V, t + i_{\{x\}})$.

Consider the augmented diagram $D' = (D_0 + 1_{\{x\}}, D_1 + f)$ of shape Q' .



Warning / Feature

- ▶ The augmented diagram D' need not commute.
 - We must check the triangle for each edge individually.

Intuition

- ▶ Each edge of the shape Q is a constraint on the *legs* of each cone C .
- ▶ Edges from Q are *not* constraints on the diagram D .

Limit and Colimit: Definition

Cones for a diagram (D_0, D_1) form a full subcategory of cones for diagram $(D_0, \emptyset)$.

Definition

A **limit** is a terminal object in the category of cones for a diagram.

Cocones for a diagram (D_0, D_1) form a full subcategory of cocones for diagram $(D_0, \emptyset)$.

Definition

A **colimit** is an initial object in the category of cocones for a diagram.

Criteria for Limiting Cone

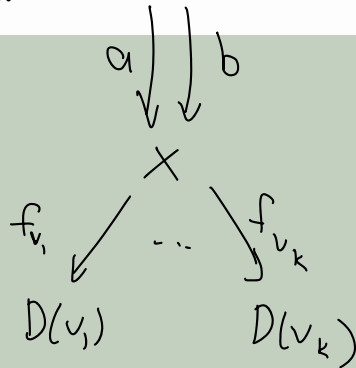
Let D be a diagram in $\mathcal{C}$ of shape $Q = (V, E, s, t)$.

Let x be a $\mathcal{C}$ -object and $f_v : x \rightarrow D(v)$ an assignment of arrows.

Theorem

$L = (x, \{f_v\}_{v \in V})$ is a limiting cone iff:

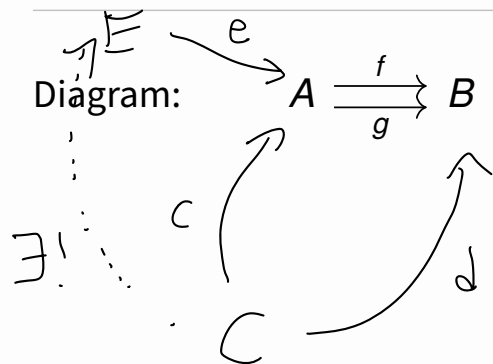
- ▶ L is a cone:
 - each triangle commutes: $f_{t(e)} = D(e) \circ f_{s(e)}$
- ▶ L is subterminal:
 - $\{f_v\}_{v \in V}$ is a “jointly monic” family
- ▶ L is weakly terminal:
 - for each other cone C , a comparison cone morphism $C \rightarrow L$ exists



$$\left[(\forall v \in V) (f_v \circ a = f_v \circ b) \right] \Rightarrow a = b$$

Limits and Colimits: Special Cases

Limit: Equalizer



$$\text{s.t. } \begin{aligned} f \circ c &= d \\ g \circ c &= d \end{aligned}$$

$$\text{equiv. } C \xrightarrow{c} A \quad \text{s.t. } f \circ c = g \circ c$$

Note

- In **Set**, the equalizer is $\{a \in A : f(a) = g(a)\} \hookrightarrow A$.

Colimit: Coequalizer

Diagram:

$$\begin{array}{ccccc} & & a & \nearrow & Q \\ A & \xrightleftharpoons[g]{f} & B & \searrow h & \downarrow \exists! \\ & & & & K \end{array} \quad \text{s.t.} \quad h \circ f = h \circ g$$

Note

- In **Set**, the coequalizer is $B \xrightarrow{[-]} B/R$ where:
 - R is the least equivalence generated by $\{\langle f(a), g(a) \rangle : a \in A\}$.

Initial Object and Terminal Object

colimit

limit

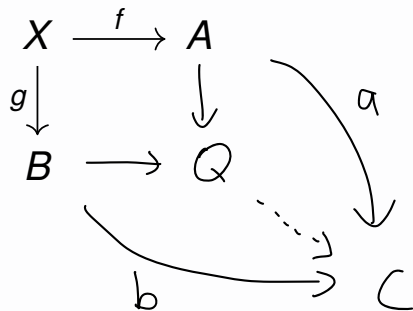
Diagram: (empty)

Intuition

- ▶ The category of cones is “isomorphic” to the original category.
- ▶ Dually, the category of cocones is “isomorphic” to the original category.

Colimit: Pushout

Diagram:

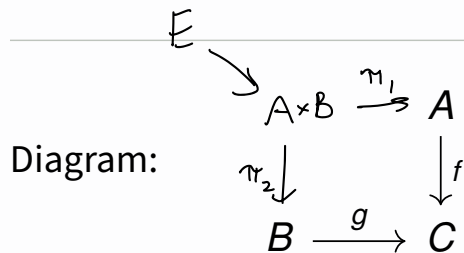


Cocores satisfy $a \circ f = b \circ g$

Note

- In **Set**, the pushout is $A \xrightarrow{[i_A(-)]} (A + B)/R \xleftarrow{[i_B(-)]} B$ where:
 - R is the least equivalence generated by $\{\langle f(x), 0 \rangle, \langle g(x), 1 \rangle : x \in X\}$.

Limit: Pullback



Notes

- ▶ In **Set**, the pullback is $A \xleftarrow{\pi_1} A_{f \times g} B \xrightarrow{\pi_2} B$ where:
 - $A_{f \times g} B = \{ \langle a, b \rangle \in A \times B : f(a) = g(b) \}$
- ▶ A more common (but vaguer) notation is $A \times_C B$.

Intuition

This is arguably the most important concept in a topos!

Size of a (Co)Limit

Let $D = (D_0, D_1)$ be a diagram in $\mathcal{C}$ of shape $Q = (V, E, s, t)$.

Definition

- ▶ D is **finite** if both V and E are finite.
- ▶ D is **small** if both V and E are small.
 - (reminder: a collection is small if it is set-sized)
- ▶ D is **large** if either V or E is large.
 - (reminder: a collection is large if it is class-sized)

Definition

A (co)limit for D is called **finite** (resp. **small**) (resp. **large**) if D is respectively so.

Completeness and Cocompleteness

Let $\mathcal{C}$ be a category.

Definition

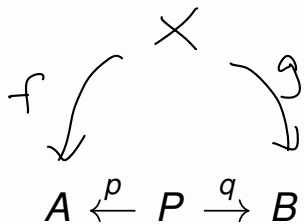
- ▶ $\mathcal{C}$ is **finitely (co)complete** if $\mathcal{C}$ has (co)limits of all finite diagrams.
- ▶ $\mathcal{C}$ is **(co)complete** if $\mathcal{C}$ has all (co)limits of all small diagrams.

Theorem / Warning

- ▶ If $\mathcal{C}$ is finite and has all finite limits, then $\mathcal{C}$ is a preorder.
- ▶ If $\mathcal{C}$ is small and has all small limits, then $\mathcal{C}$ is a preorder.
- ▶ If $\mathcal{C}$ has all large limits, then $\mathcal{C}$ is a preorder.

(Proof is out of scope for this intro.)

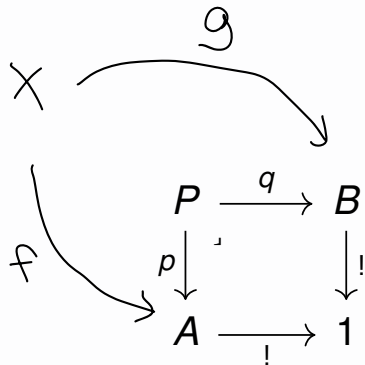
Pullbacks over Terminal Objects are Products



is a product

categories of
cones are iso

vs.



s.t.

$$!_A \circ f = !_B \circ g$$

always true

Equalizers Are Expressible from Products and Pullbacks

$$E \xrightarrow{e} A \rightrightarrows^f_g B$$

is an equalizer

vs.

$$\begin{array}{ccc} E & \xrightarrow{e} & A \\ e \downarrow & \lrcorner & \downarrow \langle 1_A, g \rangle \\ A & \xrightarrow{\langle 1_A, f \rangle} & A \times B \end{array}$$

vs.

$$\begin{array}{ccc} E & \xrightarrow[f \circ e]{g \circ e} & B \\ e \downarrow & \lrcorner & \downarrow \langle 1_B, 1_B \rangle \\ A & \xrightarrow{\langle f, g \rangle} & B \times B \end{array}$$

Limits Are Expressible from Equalizers and Products

Let $D = (D_0, D_1)$ be a diagram in $\mathcal{C}$ of shape $Q = (V, E, s, t)$.

Suppose the following two products exist:

$$\left(\prod_{v \in V} D_0(v), \{\pi_v\}_{v \in V} \right) \quad \text{and} \quad \left(\prod_{e \in E} D_0(t(e)), \{\pi'_e\}_{e \in E} \right)$$

Theorem

D has a limit iff an equalizer for the following exists:

$$\prod_{v \in V} D_0(v) \begin{array}{c} \xrightarrow{\langle D_1(e) \circ \pi_{D_0(s(e))} \rangle_{e \in E}} \\ \xrightarrow{\langle \pi_{D_0(t(e))} \rangle_{e \in E}} \end{array} \prod_{e \in E} D_0(t(e))$$

(Proof is out of scope for this intro.)

Finite Limits Are Expressible From Pullbacks And Terminal Object

Corollary

If $\mathcal{C}$ has pullbacks and a terminal object, then $\mathcal{C}$ is finitely complete.

Pullbacks

Pullbacks in Set

Theorem

Let $f : A \rightarrow C$ and $g : B \rightarrow C$ be functions. The following is a pullback:

$$\begin{array}{ccc} \bigcup_{c \in C} f^{-1}(c) \times g^{-1}(c) & \xrightarrow{\pi_1} & A \\ \pi_2 \downarrow & & \downarrow f \\ B & \xrightarrow{g} & C \end{array}$$

Example: Inverse Image

Theorem

If $f : A \rightarrow B$ is a function and $T \subseteq B$, then

$$\begin{array}{ccc} f^{-1}(T) & \xrightarrow{f} & T \\ \downarrow & & \downarrow i \\ A & \xrightarrow{f} & B \end{array} \text{ is a pullback.}$$

$$b \in B$$

$$f^{-1}(b) \times \underbrace{i^{-1}(b)}_{\subseteq \{b\}} \cong \bigcup_{b \in T} f^{-1}(\{b\}) = f^{-1}(T)$$

Example: Pullback for Product Projections

Theorem

$$\begin{array}{ccc} A \times X & \xrightarrow{\pi_1} & A \\ f \times 1_X \downarrow & & \downarrow f \\ B \times X & \xrightarrow{\pi_1} & B \end{array} \text{ is a pullback.}$$

$$\begin{array}{c} \pi_2 \downarrow \\ X \end{array}$$

Lemma: Pullback of Monic

Theorem

If $\begin{array}{ccc} A & \xrightarrow{k} & B \\ g \downarrow & & \downarrow f \\ C & \xrightarrow{h} & D \end{array}$ is a pullback and f is monic, then g is also monic.

$$\{g, k\} \text{ is j.m.} \Rightarrow \{g, f \circ k\} \text{ is j.m.} \Rightarrow \{g, h \circ g\} \text{ is j.m.} \Rightarrow \{g, g\} \text{ is j.m.}$$

Intuition

- ▶ A conceptual way to prove this involves reasoning with jointly monic families:
 - They can be “grafted”.
 - They can be “truncated” (but not at the root).

$\Downarrow$
 $\{g\} \text{ is j.m.}$

Pullback of Monic: Partial Converse

Theorem

Suppose $\begin{array}{ccc} A & \xrightarrow{g} & B \\ \downarrow n & & \downarrow m \\ C & \xrightarrow{f} & D \end{array}$. Suppose $\begin{array}{ccc} X & \xrightarrow{\forall q} & B \\ \downarrow \forall p & & \downarrow m \\ C & \xrightarrow{f} & D \end{array}$ implies $\begin{array}{ccc} X & \overset{\exists x}{\dashrightarrow} & A \\ \searrow p & & \swarrow n \\ & C & \end{array}$.

Then, the first square is a pullback.

Pullback of Monic: Corollary to Partial Converse

Corollary

If $\begin{array}{ccc} A & \xrightarrow{g} & B \\ n \downarrow \cong & \nearrow \tilde{n}^{-1} & \downarrow m \\ C & \xrightarrow{f} & D \end{array}$ commutes with monic m and iso n , then it is a pullback.

Special cases:

$$\begin{array}{ccc} A & \xrightarrow{\cong} & B \\ \downarrow & \lrcorner & \downarrow \\ C & \xrightarrow{\cong} & D \end{array}$$

$$\begin{array}{ccc} C & \xlongequal{\quad} & C \\ \parallel & \lrcorner & \downarrow m \\ C & \xrightarrow{m} & D \end{array}$$

Pullback Characterization of Monic

Theorem

If $\begin{array}{ccc} A & \xrightarrow{1_A} & A \\ 1_A \downarrow & \lrcorner & \downarrow m \\ A & \xrightarrow{m} & B \end{array}$ is a pullback, then m is monic.

Lemma: Product of Pullback Squares

Theorem

$$\begin{array}{ccc} A_1 & \xrightarrow{p_1} & B_1 \\ q_1 \downarrow & \lrcorner & \downarrow f_1 \\ C_1 & \xrightarrow{g_1} & D_1 \end{array} \quad \text{and} \quad \begin{array}{ccc} A_2 & \xrightarrow{p_2} & B_2 \\ q_2 \downarrow & \lrcorner & \downarrow f_2 \\ C_2 & \xrightarrow{g_2} & D_2 \end{array} \quad \text{imply} \quad \begin{array}{ccc} A_1 \times A_2 & \xrightarrow{p_1 \times p_2} & B_1 \times B_2 \\ q_1 \times q_2 \downarrow & \lrcorner & \downarrow f_1 \times f_2 \\ C_1 \times C_2 & \xrightarrow{g_1 \times g_2} & D_1 \times D_2 \end{array}$$

Lemma: Pullback With “Diagonal”

Theorem

If m is monic, then

$$\begin{array}{ccc}
 A & \xrightarrow{\langle 1_A, 1_A \rangle} & A \times A \\
 m \downarrow & \lrcorner & \downarrow m \times m \\
 B & \xrightarrow{\langle 1_B, 1_B \rangle} & B \times B
 \end{array}$$

is a pullback.

$$\begin{array}{ccc}
 A & = & A \\
 \parallel \lrcorner & & \downarrow m \\
 A & \xrightarrow{m} & B
 \end{array}$$

$$\begin{array}{ccc}
 A \times A & = & A \times A \\
 \parallel \lrcorner & & \downarrow m \times m \\
 A \times A & \xrightarrow{m \times m} & B \times B
 \end{array}$$

Pullback Projections Can Coincide

Theorem

If
$$\begin{array}{ccc} E & \xrightarrow{e} & A \\ e \downarrow & \lrcorner & \downarrow f \\ A & \xrightarrow{g} & B \end{array}$$
 is a pullback, then e is the equalizer of $f, g : A \rightrightarrows B$.

Note

The converse need not hold!

Lemma: Two Pullbacks Lemma

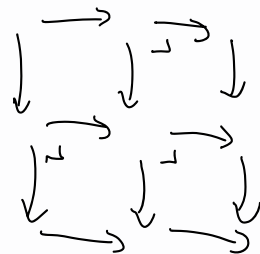
Lemma

Suppose that

$$\begin{array}{ccccc} A & \xrightarrow{f'} & B & \xrightarrow{g'} & C \\ a \downarrow & & b \downarrow & & c \downarrow \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z \end{array}$$

commutes and that the right square is pullback.

Then, the left square is a pullback iff the outer rectangle is a pullback.



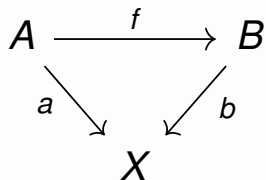
Subobject Classifier

Important Construction: Slice Category

Let $\mathcal{C}$ be a category. Let X be a $\mathcal{C}$ -object.

Definition

- The **slice category** over X is denoted $\mathcal{C}/X$.
 - An object is an ordered pair $(A \in \mathcal{C}_0, a : A \rightarrow X)$.
 - An arrow from (A, a) to (B, b) is an $f : A \rightarrow B$ such that $a = b \circ f$.



depicts an arrow f from (A, a) to (B, b) .

Slice Category vs. Comma Category

Warning

- ▶ In categorical lingo, the term “comma category” is usually reserved for a much more general construction than what Goldblatt proposes.
- ▶ The term “slice category” or “overcategory” is often used to refer to $\mathcal{C}/X$.

Theorem

If $\mathcal{C}$ is a topos, then $\mathcal{C}/X$ is a topos as well.

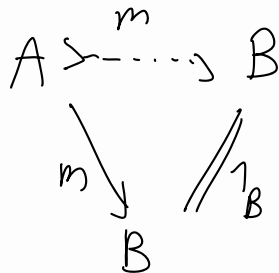
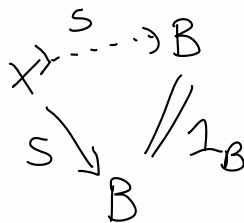
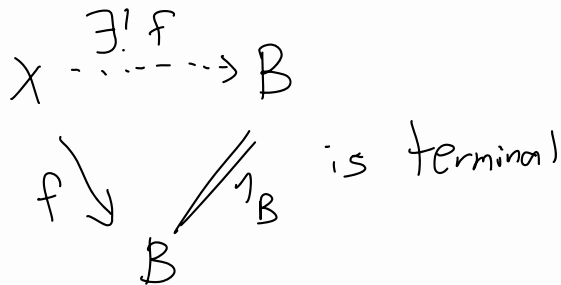
(Proof is out of scope for this intro.)

Monic: Equivalent Characterization

Let $\mathcal{C}$ be a category.

Theorem

A monic $m : A \rightarrow B$ in $\mathcal{C}$ is the same as a subterminal (A, m) in a slice category $\mathcal{C}/B$.



Subobject: Definition

Definition

A **subobject** of X is an isomorphism class of subterminals in the slice category $\mathcal{C}/X$.



$$[m: M \rightarrow X] = [m': M' \rightarrow X]$$

Intuition

- ▶ A subobject of X describes a portion of X .
- ▶ A subobject of a terminal describes an external truth value.

Example: Subobjects in Set

$$M \cong \{x \in X : (\exists a)(m(a) = x)\}$$

The diagram illustrates the relationship between a set M and its image X under a function m . The set M is isomorphic to the set of elements x in X that are in the image of m . The function m maps elements of M to elements of X . The set X is mapped to the set of subobjects $2 = \{0, 1\}$, where 0 represents the empty set and 1 represents the full set X . The mapping from X to 2 is defined by the characteristic function of the image of m .

$$x \mapsto \begin{cases} 0 & \text{if } x \notin \text{Im}(m) \\ 1 & \text{if } x \in \text{Im}(m) \end{cases}$$

Note

Note that $\text{Sub}(X) \cong \mathcal{P}(X) \cong 2^X$.

Lemma: Pullback of Subobject

Lemma

Pullbacks of subobjects are well-defined subobjects, independent of choices of monics. Moreover, the unspecified leg is uniquely determined (upon making those choices).

$$\begin{array}{ccccccc} M' & \xrightarrow{\cong} & M & \dashrightarrow & N & \xrightarrow{\cong} & N' \\ m' \downarrow & & m \downarrow & & n \downarrow & & n' \downarrow \\ X & \xlongequal{\quad} & X & \xrightarrow{f} & Y & \xlongequal{\quad} & Y \end{array}$$

Lemma: Composing Pullback Squares between Subobjects

Lemma

Let $[a : A \hookrightarrow X]$ and $[b : B \hookrightarrow Y]$ and $[c : C \hookrightarrow Z]$ be subobjects.

Let $[a]$ be a pullback of $[b]$ along $f : X \rightarrow Y$.

Let $[b]$ be a pullback of $[c]$ along $g : Y \rightarrow Z$.

Then, $[a]$ is a pullback of $[c]$ along $g \circ f : X \rightarrow Z$.

$$\begin{array}{ccccc} A & \dashrightarrow & B & \dashrightarrow & C \\ a \downarrow & & b \downarrow & & c \downarrow \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z \end{array}$$

Important Construction: $\text{el}(\text{Sub})$

Let $\mathcal{C}$ be a category with pullbacks.

Theorem

- ▶ There is a category, denoted $\text{el}(\text{Sub}_{\mathcal{C}})$.
 - An object is an ordered pair (X, s) where s is a subobject of X .
 - An arrow from $(X, [m])$ to $(Y, [n])$ is an $f : X \rightarrow Y$ that fits in a pullback square:

$$\begin{array}{ccc} M & \dashrightarrow & N \\ m \downarrow & \lrcorner & \downarrow n \\ X & \xrightarrow{f} & Y \end{array}$$

Subobject Classifier: Definition?

Let $\mathcal{C}$ be a category with pullbacks.

Definition

- ▶ A **subobject classifier** in $\mathcal{C}$ is a terminal object of $\text{el}(\text{Sub}_{\mathcal{C}})$.
 - A priori, “the” terminal object is denoted $(\Omega, [true : \Omega_0 \twoheadrightarrow \Omega])$.

Domain of *true* is Terminal

Let $\mathcal{C}$ be a category with pullbacks.

Theorem

Let $(\Omega, [true : \Omega_0 \rightarrow \Omega])$ be a subobject classifier in $\mathcal{C}$. Then Ω_0 is a terminal object.

$$\begin{array}{ccc}
 X & \xrightarrow{\quad} & \Omega_0 \\
 \downarrow \eta_X & \lrcorner & \downarrow true \\
 X & \xrightarrow{\exists!} & \Omega
 \end{array}$$

$$\begin{array}{ccccc}
 X & \xrightarrow{f} & \Omega_0 & \xrightarrow{\quad} & \Omega_0 \\
 \parallel & \lrcorner & \parallel & \lrcorner & \downarrow true \\
 X & \xrightarrow{g} & \Omega_0 & \xrightarrow{true} & \Omega
 \end{array}$$

Corollary

If $\mathcal{C}$ has a subobject classifier, then $\mathcal{C}$ has finite limits.

$$\begin{aligned}
 true \circ f &= true \circ g \\
 \therefore f &= g
 \end{aligned}$$

Warning: *true* is **Not** Determined by Domain and Codomain

Let $\mathcal{C}$ be a category with pullbacks.

Theorem / Warning

Let $(\Omega, [true : \Omega_0 \twoheadrightarrow \Omega])$ be a subobject classifier in $\mathcal{C}$. Then for each terminal object T ,

$$\mathcal{C}(T, \Omega) \cap [true : \Omega_0 \twoheadrightarrow \Omega] \cong \text{Aut}(\Omega)$$

Example in Set

- ▶ Both $0 \mapsto 0 : \{0\} \rightarrow \{0, 1\}$ and $0 \mapsto 1 : \{0\} \rightarrow \{0, 1\}$ are subobject classifiers.
 - The convention is to make $true(0) = 1$
 - (It's the image of “true” that matters more than the domain of “true”)

Subobject Classifier: Definition (Revisited)

Let $\mathcal{C}$ be a category with finite limits.

Definition

- ▶ A **subobject classifier** is a terminal object of $\text{el}(\text{Sub}_{\mathcal{C}})$.
 - A fortiori, “the” terminal object of $\text{el}(\text{Sub}_{\mathcal{C}})$ is denoted $(\Omega, \text{true} : 1 \rightarrow \Omega)$.

Ω -Axiom

Characteristic Arrow: Definition

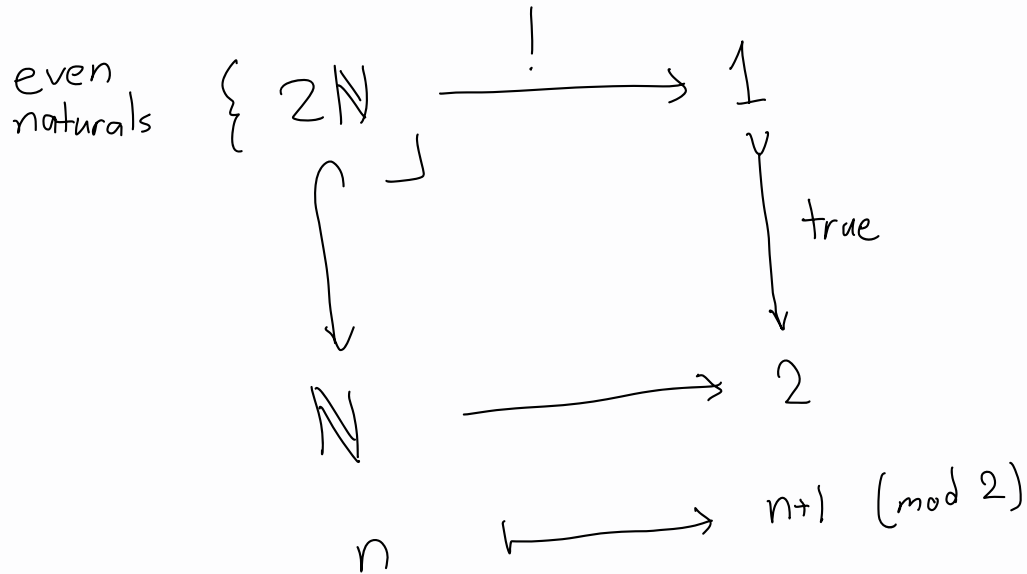
Let $\mathcal{C}$ be a category with pullbacks. Let $(\Omega, \text{true} : 1 \rightarrow \Omega)$ be a subobject classifier. Let $m : S \rightarrow X$ be a monomorphism.

Definition

The **characteristic arrow** of m is the (unique) arrow χ_m in $\text{el}(\text{Sub}_{\mathcal{C}})$ from $[m]$ to $[\text{true}]$.

$$\begin{array}{ccc} S & \dashrightarrow & 1 \\ m \downarrow & \lrcorner & \downarrow \text{true} \\ X & \xrightarrow{\chi_m} & \Omega \end{array}$$

Characteristic Arrow: Example



Exercise Review

Outro

Reading for Next Lecture

- ▶ Minimal
 - Sections 7 and 8 of chapter 4
 - Sections 1 and 2 of chapter 5
- ▶ Excellent
 - Sections 5-8 of chapter 4
 - Sections 1-3 of chapter 5
- ▶ Extra
 - First 11 videos of this playlist